AI LAB 7.py - C:/Users/K Aishwarya/Desktop/AI/AI LAB 7.py (3.14.2)

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```
    print("BREADTH-FIRST SEARCH")
    print("--------------------")
    print("Starting Node:", start)
    print("\nTraversal Order:")

    while queue:
        node = queue.popleft()

        if node not in visited:
            print("Visited:", node)
            visited.add(node)

            for neighbour in graph[node]:
                if neighbour not in visited:
                    queue.append(neighbour)

    print("\nFinal BFS Traversal:", " -> ".join(visited_order))

visited_order = []

def bfs(graph, start):
    visited = set()
    queue = deque([start])
    visited_order.clear()

    print("BREADTH-FIRST SEARCH")
    print("--------------------")
    print("Starting Node:", start)

    while queue:
        node = queue.popleft()

        if node not in visited:
            visited.add(node)
            visited_order.append(node)

            print("Visited Node:", node)

            for neighbour in graph[node]:
                if neighbour not in visited:
                    queue.append(neighbour)

    print("\nFinal BFS Traversal:", " -> ".join(visited_order))


bfs(graph, 'A')
```

```
>>>
    ============================================================
    BREADTH-FIRST SEARCH
    --------------------
    Starting Node: A
    Visited Node: A
    Visited Node: B
    Visited Node: C
    Visited Node: D
    Visited Node: E
    Visited Node: F
    Visited Node: G

    Final BFS Traversal: A -> B -> C -> D -> E -> F -> G
>>>
```